

Unit III
COMPUTER FUNDAMENTALS

Functional Units of a Digital Computer: Von Neumann Architecture – Operation and
Operands of Computer Hardware Instruction – Instruction Set Architecture (ISA): Memory
Location, Address and Operation – Instruction types– Addressing Modes

Course Objective – COB1: To analyze Functions, memory and addressing modes of computer.
Course Outcome - CO1 : Gain Complete knowledge about functional unit and memory.

Bloom's Taxonomy Level: K1,K2,K3,K4

Part A

S.No.	Question	BTL*
1.	What are the basic functional units of a computer? Ans: A computer consists of five functionally independent main parts namely -Input Unit -Memory Unit -Arithmetic and logic Unit -Output Unit -Control Unit	K1
2.	Define RAM. Ans: Memory in which any location can be reached in a short and fixed amount of time after specifying its address is called random access memory.	K1
3.	Define addressing mode.	K1

	The different ways in which the location of an operand is specified in an instruction are referred to as addressing modes.	
4.	What are the different types of addressing modes available? Ans: The different types of addressing modes available are: -Immediate addressing mode -Register addressing mode -Direct or absolute addressing mode -Indirect addressing mode -Indexed addressing mode -Relative addressing mode -Auto increment -Auto decrement	K2
5.	What is indirect addressing mode? Ans: The effective address of the operand is the contents of a register or memory Location whose address appears in the instruction What is indexed addressing mode? Ans: The effective address of the operand is generated by adding a constant value to the contents of a register.	K2
6.	Define auto increment mode of addressing? Ans: The effective address of the operand is the contents of a register specified in the instruction. After accessing the operand, the contents of this register are automatically incremented to point to the next item in the list. Define auto decrement mode of addressing? Ans: The contents of a register specified in the instruction are first automatically decremented and are then used as the effective address of the operand.	K1
7.	What is index register? In index mode the effective address of the operand is generated by adding a constant value to the contents of a register. The register used	K1

	may be either a special register or may be any one of a set of general purpose registers in the processor. This register is referred to as an index register.	
8.	What do you mean by relative addressing mode? Ans: The effective address is determined by the index mode using the program counter in place of the general purpose register Ri.	K1
9.	What is Stack? Ans: A stack is a list of data elements, usually words or bytes with the accessing restriction that elements can be added or removed at one end of the list only. It follows last in first out (LIFO) mechanism. What is a queue? Ans: Is a type of data structure in which the data are stored in and retrieved on a First in first out(FIFO) basis. It grows in the direction of increasing addresses in the memory. New data are added at the back (High-address end) and retrieved from the front (low-address end) of the queue.	K1
10.	What do you mean by assembler directives? Ans: These are the instructions which direct the program to be executed. They have no binary equivalent so they are called pseudo-opcodes. These instructions are used to define symbols, allocate space for variable, generate fixed tables etc. Examples : END, NAME	K1
11	What is assembly language? A complete set of symbolic names and rules for the use of machines constitute a programming language, generally referred to as an assembly language.	K1
12.	What is an interrupt? Ans: An interrupt is a request from an I/O device for service by the processor. The processor provides the requested service by executing an appropriate interrupt service routine.	K1
13.	Explain about Bus. Ans: Bus is a group of lines that serves as a connecting path for several devices. In addition to the lines that carry the data , the bus must have	K1

	the lines for address and control purposes.	
14.	What do you mean by multiprogramming or multitasking? Ans: The operating system manages the concurrent execution of several application programs to make best possible use of computer resources. This pattern of concurrent execution is called multiprogramming or multitasking.	K3
15.	What is instruction register (IR) and program counter (PC) used for Ans: The instruction register (IR) holds the instruction that is Currently being executed .Its output is available to the control circuits which generate the timing signals that control the various processing elements. The program counter PC is used to keep track of the execution of the program. It contains the memory address of the next instruction to be Fetched and executed.	K1
16.	Define memory access time. Ans: The time required to access one word is called memory access time.	K4
17.	Difference between Primary and Secondary Storage System?	K2

Sr.No.	Primary Memory	Secondary Memory
1.	Primary memory is temporary.	Secondary memory is permanent
2.	Primary memory is directly accessible by Processor/CPU.	Secondary memory is not directly accessible by the CPU.
3.	Nature of Parts of Primary memory varies, RAM- volatile in nature. ROM- Non-volatile.	It's always Non-volatile in nature.
4.	Primary memory devices are more expensive than secondary storage devices.	Examples: RAM, ROM, Cache memory, PROM, EPROM, Registers, etc.

Part B

S.No.	Question	BTL *
1.	Explain different functional units of digital computer.	K3
2.	Draw and Explain the Von Neumann architecture	K3
3.	Write a note on Operands and Operation of computer Hardware	K3
4.	What is ISA and Explain the MIPS Instruction Set Architecture	K3
5.	Explain the Data Transfer operation between memory and processor	K4
6.	Define addressing modes and Elaborate the different types of Addressing modes.	K3
7.	Explain briefly about memory.	K2
8.	Explain briefly about 1. Byte Addressability 2. Big-endian Assignment 3. Little-endian Assignment	K3